

LLM-enhanced Career Knowledge Graph Understanding for Job Mobility Prediction

SHUTING CUI, Information Hub, The Hong Kong University of Science and Technology - Guangzhou Campus, Guangzhou, China

YING SUN*, Information Hub, The Hong Kong University of Science and Technology - Guangzhou Campus, Guangzhou, China

YUTING ZHANG, Information Hub, The Hong Kong University of Science and Technology - Guangzhou Campus, Guangzhou, China

QINGXIN MENG, Department of Entrepreneurship, Marketing and Management Systems, University of Nottingham Ningbo China, Ningbo, China

HENGSHU ZHU*, Computer Network Information Center Chinese Academy of Sciences, Haidian, China and University of the Chinese Academy of Sciences, Beijing, China

Job mobility prediction is an emerging field with significant benefits for organizations and individuals, including enhanced job recommendations and career planning. Existing approaches leveraging textual information primarily rely on traditional neural network embedding methods, which are limited in capturing relationships between job-related entities at the semantic level. While Large Language Models (LLMs) offer greater potential by utilizing extensive background knowledge to better understand latent patterns in career trajectories, pure LLMs often struggle with hallucination issues due to insufficient domain-specific knowledge. To bridge this gap, we propose a knowledge graph-enhanced LLM framework, **Smart Career Predictor (SCP)**. Our approach introduces three key innovations: (1) a job-related sub-knowledge graph retrieval mechanism to retain the most relevant job transition patterns, addressing the hallucination problem by providing enriched background knowledge; (2) an entropy-based retrieval score to alleviate graph imbalance issues by ensuring a more equitable representation of career entities; and (3) a multi-modal adapter to effectively integrate textual and graph-based information, enabling more accurate and context-aware job mobility predictions. Experimental results on two real-world career datasets show that our model significantly outperforms state-of-the-art baselines in next career hop prediction. The code for this paper is available at <https://github.com/cuishuting/SCP>.

CCS Concepts: • **Information systems** → **Data mining**.

Additional Key Words and Phrases: Job Mobility Prediction, Knowledge Graph, LLM

1 Introduction

Understanding job transitions is critical for both employers to attract and retain top talent, and for individuals to search suitable positions and advance their careers. For employers, accurately modeling job transitions helps recruiters identify and attract candidates whose career trajectories align with the organization's strategic talent needs [30]. For individuals, analyzing career mobility through data-driven models reveals how factors such as skills,

*Corresponding author

Authors' Contact Information: Shuting Cui, Information Hub, The Hong Kong University of Science and Technology - Guangzhou Campus, Guangzhou, Guangdong, China; e-mail: scui847@connect.hkust-gz.edu.cn; Ying Sun, Information Hub, The Hong Kong University of Science and Technology - Guangzhou Campus, Guangzhou, Guangdong, China; e-mail: yings@hkust-gz.edu.cn; Yuting Zhang, Information Hub, The Hong Kong University of Science and Technology - Guangzhou Campus, Guangzhou, Guangdong, China; e-mail: y Zhang755@connect.hkust-gz.edu.cn; Qingxin Meng, Department of Entrepreneurship, Marketing and Management Systems, University of Nottingham Ningbo China, Ningbo, Zhejiang, China; e-mail: Qingxin.Meng@nottingham.edu.cn; Hengshu Zhu, Computer Network Information Center Chinese Academy of Sciences, Haidian, China and University of the Chinese Academy of Sciences, Beijing, Beijing, China; e-mail: zhuhengshu@gmail.com.

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job satisfaction, and industry growth influence career progression and talent development [21]. With the rise of online professional networks (OPNs) like LinkedIn, the accessibility of vast digital resumes has fueled the popularity of data-driven approaches to job mobility prediction [24, 27, 49]. The existing body of research on job mobility prediction can be broadly categorized into three distinct approaches: those that primarily utilize textual information derived from historical job transitions and user profile data [24, 27], those that focus on the graphical structures formed by historical job transitions [49], and those that integrate both textual and graphical information for a more comprehensive analysis [46]. However, these approaches leveraging textual information primarily rely on traditional neural network embedding methods, such as Word2Vec or Doc2Vec. This limits their ability to understand relationships between job-related entities beyond the semantic level. In practice, relationships among career entities (e.g., job title, company, and skill) are often hierarchical, contextual, and causally linked rather than purely semantic. For instance, the transition from “Data Analyst” at a finance firm to “Product Manager” in a tech company cannot be inferred simply from text similarity, but requires understanding latent dependencies such as domain adaptability, skill transferability, and organizational role evolution. A deeper representation of such inter-entity relations is therefore essential for modeling realistic career mobility patterns and long-term progression. In contrast, the advent of Large Language Models (LLMs) brings greater potential for understanding historical job path information. By encoding both explicit textual semantics and implicit relational cues learned from large-scale knowledge, LLMs can reason about entity interactions, contextual dependencies, and role transitions beyond surface-level co-occurrences. By learning from extensive and diverse background knowledge, LLMs can better capture the latent information and relationship embedded in an individual’s educational background, personal descriptions, and job experiences, leading to more accurate predictions of the next job or company. The existing research has not yet fully unlock this potential.

While LLMs are powerful tools available for job move prediction, they also face challenges, particularly the issue of hallucination, where the model generates information that is not grounded in the input data or reality. This problem becomes more pronounced when LLMs lack sufficient domain knowledge to guide the generation of accurate and contextually relevant answers. To alleviate the hallucination problem of LLMs, incorporating job transition-related knowledge graphs (KGs) as prompts can be an effective approach. KGs provide structured, domain-specific information that can guide the model’s predictions [28, 41]. Some recent research solve this problem by retrieving sub-KG given query-related entities [13, 38]. However, the retrieved sub-KGs often include entities that appear more frequently in the entire KG. Such an imbalance in the retrieved sub-KG can result in skewed predictions, as the model might prioritize popular or frequently connected entities over others that are relevant to the specific context. Therefore, addressing this issue requires strategies to ensure that the sub-KG retrieval process captures a more balanced and relevant representation of the knowledge graph.

To address the above two mentioned research gaps in a unified framework, we propose a novel LLM-enhanced career KG understanding approach called **SCP** for next job mobility prediction. We firstly build the career KG from massive career trajectories to represent the entire career transition patterns. For each user, we firstly extract the sub-KG based on the k-hop neighbors from the entire KG by user’s historical job experience. Then, we propose a heterogeneous graph neural network (HGNN) designed to learn embeddings for career-related entities (i.e., jobs and companies) through a link prediction task. Those job-related entity embeddings will be further utilized as background knowledge to address the hallucination problem in LLMs. Furthermore, we design an entropy-based “retrieval score” to assign more appropriate weights to the entities within the sub-KG, thereby alleviating the imbalanced sub-KG retrieval problem mentioned earlier. In addition to leveraging graph-based sub-KG representations, we integrate textual representations of job experience obtained through a BERT-based embedding framework. To effectively fuse the two modalities—graph-based and text-based representations—we propose a bi-directional cross-modality interaction adapter. This adapter facilitates comprehensive integration of the two information sources, producing a soft prompt that is enriched with enhanced background knowledge. Subsequently, we utilize quantized-LoRA [6] to fine-tune the LLM, which is then employed for job mobility prediction.

To sum up, the contributions of this work are as follows:

- We propose addressing the job mobility prediction problem by combining LLMs and career knowledge graphs to better leverage the rich textual information in users’ profiles and mitigate the hallucination problem of LLMs.
- We introduce a novel retrieval score combined with quantized LoRA technique that unifies the knowledge retrieval and parameter-efficient-fine-tuning LLM inference into an end-to-end learnable framework.

- We develop an innovative bi-directional cross-modality interaction adapter that aligns text-modal and graph-modal embeddings to smoothly combine the external enhanced knowledge into LLM’s prompt to further enhance the next career hop prediction accuracy.
- Extensive experiments on two real-world career datasets demonstrate the effectiveness of our approach compared to state-of-the-art baselines in the context of job mobility prediction.

The paper is organized as follows: Section 2 reviews related work on AI-driven talent analytics, job mobility prediction and recent advances in integrating graphs with LLMs. Section 3 presents the problem formulation. Section 4 introduces the proposed method in detail. Section 5 outlines the experimental setup and discusses the results. Finally, Section 6 concludes the paper.

2 Related Work

This section overviews the related existing works in the fields of AI-driven talent analytics, job mobility prediction and LLM-based graph modeling.

2.1 AI-driven Talent Analytics

In the last decade, talent analytics has emerged as a promising field in applied data science for human resource management [36], garnering significant attention from AI communities and inspiring numerous research efforts [31, 37]. In the era of big data, large-scale talent data enables organizations to make quantitative talent-related decisions, providing insights into talent patterns and management for effective decision-making in fast-evolving business environments [34, 35]. Generally, research in talent analytics can be broadly categorized into **talent management**, **organization management**, and **labor market analysis**. To be specific, **talent management** aids in attracting, training, and retaining high-potential employees, enabling practical talent matching [32, 47] and predicting employee performance or turnover [23]; **organization management** leverages relationship data, such as organizational structure and communication patterns, to diagnose organizational health [48], optimize teams [43], and measure performance from a macro-level perspective, **labor market analysis** examines talent demands to craft effective recruitment and organizational strategies [51, 52]. Building upon the advancements in AI-driven talent analytics, our work integrates talent management and labor market analysis by leveraging macro-level transition patterns from career knowledge graphs (KGs) to enhance practical talent job mobility prediction.

2.2 Job Mobility Prediction

Job mobility prediction is a long-standing research topic in human resource management [45]. Prior work has utilized data mining techniques and traditional deep learning models in the job mobility prediction area. Li et al. [24] proposed a contextual LSTM model to integrate the profile context and career path dynamics simultaneously for predicting the next company and position of talents. Meng et al. [27] proposed a hierarchical neural network to integrate three levels of individual information for job mobility prediction. Zhang et al. [49] proposed a novel model to combine the heterogeneous graph neural network and sequential dual-GRU framework to involve macro-level job transition patterns into job mobility prediction. Yamashita et al. [46] proposed a framework called NAOMI, which includes multi-view graph embeddings, BERT embeddings, job duration masking, and neural collaborative reasoning to predict one’s future career pathway as a sequence of N steps. Nevertheless, the aforementioned data-driven job mobility prediction models fail to fully leverage the rich textual information in users’ digital resumes, which is crucial for making accurate job mobility predictions. With the development of LLM, exploring how to utilize the rich career domain contextual information to enhance job mobility prediction becomes a prevalent research topic. Du et al. [9] fine tuned the backbone LLM using resume datasets and used the resulting representations as input to an occupation model. Du et al. [10] proposed an interactive resume completion method to mitigate hallucination in LLM-generated content. Besides, they design a GAN-based aligning method to refine LLM’s representations of low-quality resumes to enhance the performance for job recommendation. Chen et al. [4] investigated the potential of LLMs for job position prediction at both individual and group levels without relying on historical training data, by comparing outputs from different LLMs and prompt design perspectives using the most recent job position of the collected resumes. Different

from existing works, we investigate how to align the retrieved external knowledge from the career KG with LLM for better next career hop prediction and reducing hallucination in LLM outputs.

2.3 LLM-based Graph Modeling

The advent of Large Language Models (LLMs) has significantly transformed the research paradigm in artificial intelligence. As LLMs are applied to an ever-growing range of tasks, their ability to process complex structured data [20] and to provide transparent and interpretable reasoning [18] has become increasingly crucial. In particular, in our interconnected world, a significant portion of real-world data inherently exhibits a graph structure, as seen in the Web [3], e-commerce [17], recommendation systems [12], chemistry [50, 53] and many other domains. Recently, integrating LLMs with graph machine learning techniques has received increasing research attention [28]. Wu et al. [44] proposed a meta-path prompt constructor to encode interactions in heterogeneous behavioral graph into natural language prompts to enhance the matching accuracy between job seekers and candidate jobs. They further designed a path augmentation module including path shuffling, an adaptive path selector, and a hybrid mechanism to alleviate the prompt bias introduced by path-based sequence input. He et al. [13] proposed a general question-answering framework targeting real-world textual graphs towards multiple applications. They proposed a method called G-Retriever to perform retrieval-augmented generation (RAG) over a graph by formulating this task as a Prize-Collecting Steiner Tree optimization problem. The backbone LLM was then fine-tuned via soft prompting to better understand the retrieved subgraph. Tian et al. [38] explored enhancing pre-trained LLMs with retrieved grounded knowledge by proposing a method called: graph neural prompting (GNP), which encompasses various designs to combine GNN and LLM. Perozzi et al. [29] introduced a parameter-efficient method called GraphToken to represent structured data for LLMs by learning an encoding function that augments prompts with structural information. Unlike prior work, we treat graph and text information as two distinct modalities and explicitly model the interactions between them to better adapt the graph knowledge into LLM's prompt. Besides, we design a retrieval score that enables end-to-end learning of how graph knowledge enhances LLM inference, with the goal of reducing hallucination.

3 Preliminaries

In this section, we first present the construction of career KG. Subsequently, we formally define the job mobility prediction problem within our framework.

3.1 Career KG Construction

We conduct experiments on two real-world datasets: FIN and IT, including all the LinkedIn users in the finance and information technology industries respectively whose career experience and educational background are complete, with the length of each career trajectory ranging from two to ten and the cut-off date being December 1, 2019. We construct the Financial and IT career KG from these users' career trajectories. The career knowledge graph (KG) is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = (\mathcal{V}_c \cup \mathcal{V}_j)$ is the set of nodes, $\mathcal{E} = (\mathcal{E}_{cc} \cup \mathcal{E}_{cj} \cup \mathcal{E}_{jj} \cup \mathcal{E}_{jc})$ is the set of edges, $\mathcal{V}_c$ represents the set of companies, and $\mathcal{V}_j$ represents the set of jobs. Specifically, each edge $\{(c_h, r_{cc}, c_t)\} \in \mathcal{E}_{cc}$ represents transition relationship between company c_h and c_t ; each edge $\{(c_h, r_{cj}, j_t)\} \in \mathcal{E}_{cj}$ represents company c_h has job j_t ; each edge $\{(j_h, r_{jj}, j_t)\} \in \mathcal{E}_{jj}$ represents transition relationship between job j_h and j_t ; each edge $\{(j_h, r_{jc}, c_t)\} \in \mathcal{E}_{jc}$ represents job j_h belongs to company c_t . The notations r_{cc} , r_{cj} , r_{jj} , r_{jc} represent relationships *trans_to_comp*, *has*, *trans_to_job* and *belongs_to* respectively. The statistics of the constructed career KG can be found in Table 1. Each type of node has its own textual attributes. We store the name and the type for the company nodes and job title for the job nodes. We utilize Sentence Transformer [33] to get both the company names' and job titles' initial embeddings with a fixed size of 384.

3.2 Problem Formulation

DEFINITION 1 (Job Mobility Prediction Problem). Given a dataset $\mathcal{D} = \{(\mathcal{J}(u), \Omega(u)), u \in \mathbf{U}\}$, where $\mathcal{J}(u) = \{\mathcal{J}_1, \mathcal{J}_2, \dots, \mathcal{J}_L | u\}$ represents career trajectory for user u . $\Omega(u)$ denotes personal-specific features, including education background and self-description. $\mathcal{J}(i) = (c_i, p_i, d_i)$ indicates user u 's i -th career with company name c_i , job title p_i and career description d_i including the start and end time, the address and job content of i -th career. For a person

Table 1. Statistics of Career KG

Entities and Relationships	Finance Knowledge Graph	IT Knowledge Graph
Node Type	Total Number of Nodes	
COMPANY	5,365	7,638
JOB	8,464	7,885
Edge Type	Total Number of Edges	
Job-belongs_to-Company	23,835	26,759
Company-has-Job	23,835	26,759
Company-trans_to_comp-Company	3,405	4,526
Job-trans_to_job-Job	5,424	6,425

$u \notin \mathcal{U}$, given his/her $(\mathcal{J}(u), \Omega(u))$, our target is to predict u 's next job mobility, including next company c_{L+1} , next job p_{L+1} .

4 Proposed Method

Below, we present our proposed method, Smart Career Predictor (SCP), in four sub-sections: **(1) Sub-KG Retrieval:** This step extracts a personalized sub-KG for each user. **(2) Multi-Meta-Path Semantic-Aware Career KG Entity Representation Learning (MMP-GAT):** This step pre-trains a GNN on the sub-KG via link prediction to obtain company and job embeddings. **(3) Soft Prompt Construction:** This step integrates textual and graph embeddings to construct soft prompts for the LLM. **(4) LLM Parameter Efficient Fine-tuning (LLM PEFT):** This step fine-tunes the LLM using the constructed prompts to predict the user's next career hop. The whole figure for the proposed pipeline SCP can be found in Figure 1.

4.1 Sub-KG Retrieval

Feeding the entire career knowledge graph into the model introduces a substantial amount of irrelevant information, which can confuse representation learning and reduce computational efficiency. To mitigate this issue, we extract a personalized subgraph centered on each user's historical jobs and companies, allowing the model to focus on the information most relevant for predicting career transitions. Guided by this idea, we design a sub-KG retrieval pipeline that constructs, for each user, a local graph that preserves semantically related entities while pruning noisy or redundant ones. Specifically, we retrieve nodes within two hops of every company and job that appears in a user's career path from the original KG. If the number of candidate job nodes in the resulting subgraph exceeds a pre-defined limit k , we select the top k nodes based on the cosine similarity between their embeddings and those of the query job nodes. The same filtering strategy cannot be applied directly to company nodes because company names typically lack meaningful semantic similarity signals relative to job titles. Instead, we leverage the industry type of company nodes (e.g., Information Technology and Services, Marketing and Advertising) as a coarse-grained categorical feature. We first select company nodes that share the same type as the query companies. If their number still exceeds k , we randomly sample k nodes from them. If no same-type companies exist in the two-hop subgraph, we randomly sample k company nodes from the entire set of retrieved company nodes.

4.2 Multi-Meta-Path Semantic-Aware Career KG Entity Representation Learning: MMP-GAT

This step learns representations from the career KG as an external domain-specific knowledge source that encodes career transition patterns. The learned graph representations serve as structured prior knowledge to mitigate LLM hallucination when generating next-career recommendations. We design a heterogeneous graph neural network (HGNN) to model transition relationships between companies and jobs. We assume that career transitions typically occur between similar entities. For instance, a user is more likely to move to another company in the same industry and choose a job with similar responsibilities, rather than shift to an unrelated domain. Motivated by this, we design a representation learning framework that captures both transition and similarity relationships among career entities.

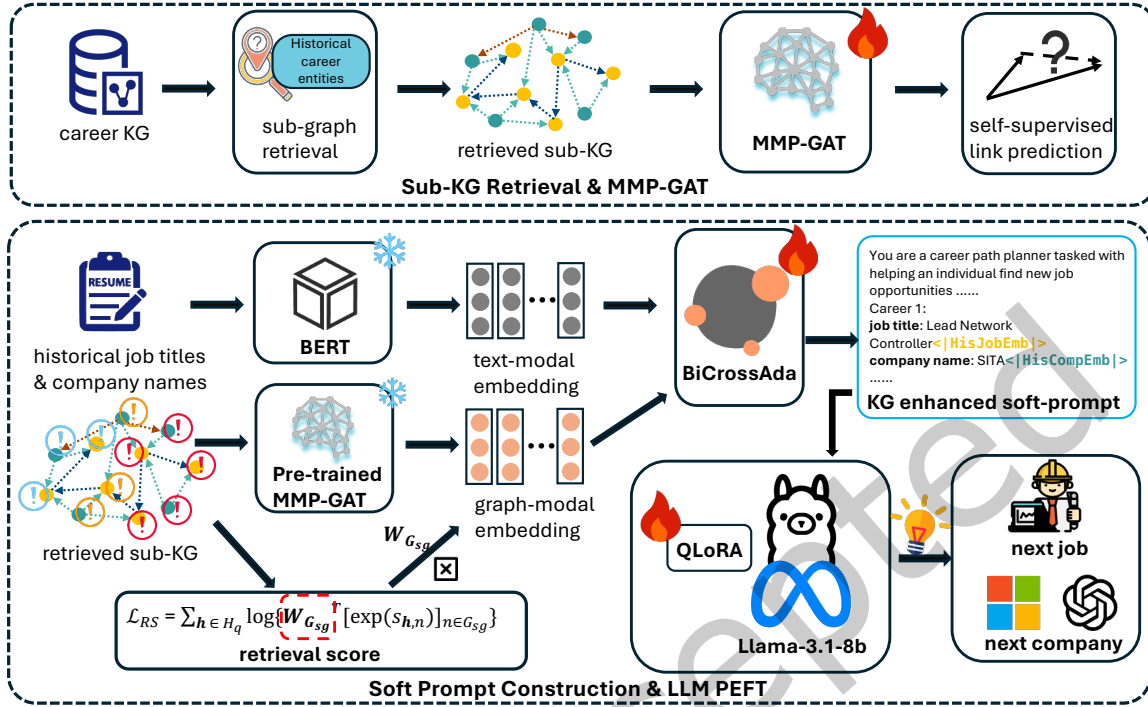


Fig. 1. Overview of the **SCP** pipeline, including: (1) sub-KG retrieval, (2) MMP-GAT pre-training for node embeddings, (3) soft prompt construction via BiCrossAda, and (4) parameter-efficient LLM fine-tuning via QLoRA.

We encode these relationships using four meta-paths, illustrated in Figure 2, where meta-paths *has-belongs_to* and *belongs_to-has* describe the similarities between company and job nodes respectively; meta-paths *has-trans_to_job-belongs_to* and *belongs_to-trans_to_comp-has* consider both the transition and similarity relationships at the same time. Since different meta-paths encode distinct semantics, nodes should attend differently to neighbors induced by different meta-paths. To achieve this goal, we design a KG entity representation learning framework called MMP-GAT: Multi-Meta-Path Semantic Graph Attention Network. For each career meta-path M_p , we apply a unique Graph Attention Layer[40] on it to aggregate information from M_p based neighbors for each node. The inputs for the framework are the company and job nodes' textual features embedding from Sentence Transformer[33], $\mathbf{h}_C = \{\mathbf{h}_{C_1}, \mathbf{h}_{C_2}, \dots, \mathbf{h}_{C_N}\} \in \mathbb{R}^{C_N \times F}$ and $\mathbf{h}_J = \{\mathbf{h}_{J_1}, \mathbf{h}_{J_2}, \dots, \mathbf{h}_{J_N}\} \in \mathbb{R}^{J_N \times F}$ where F is the static embedding size 384. For node i , we aggregate its M_p meta-path based neighbors' information performing a self-attention manner. The attention coefficients $\alpha_{ij}^{M_p}$ between node i and node j are calculated as in Equation 1, where $\mathbf{W}_{M_p} \in \mathbb{R}^{F' \times F}$ is a shared linear transformation weight matrix between $\mathbf{h}_i$ and $\mathbf{h}_j$ to obtain sufficient expressive power to transform the input features into higher-level features with embedding size F' . The operator $\parallel$ represents concatenation of two embeddings. $\mathbf{a}_{M_p} \in \mathbb{R}^{2F' \times 1}$ is the attention weight matrix.

$$\alpha_{ij}^{M_p} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}_{M_p}^T [\mathbf{W}_{M_p} \mathbf{h}_i^T \parallel \mathbf{W}_{M_p} \mathbf{h}_j^T]))}{\sum_{k \in N_{M_p}(i)} \exp(\text{LeakyReLU}(\mathbf{a}_{M_p}^T [\mathbf{W}_{M_p} \mathbf{h}_i^T \parallel \mathbf{W}_{M_p} \mathbf{h}_k^T]))}. \quad (1)$$

Then the attention coefficients $\alpha_{ij}^{M_p}$ are used to compute a linear combination of meta-path M_p based neighbors' features for each $\mathbf{h}_i$ and get the updated feature $\mathbf{h}_i^{M_p'}$ as in Equation 2:

$$\mathbf{h}_i^{M_p'} = \sigma \left(\sum_{j \in N_{M_p}(i)} \alpha_{ij}^{M_p} \mathbf{h}_j \mathbf{W}_{M_p}^T \right). \quad (2)$$

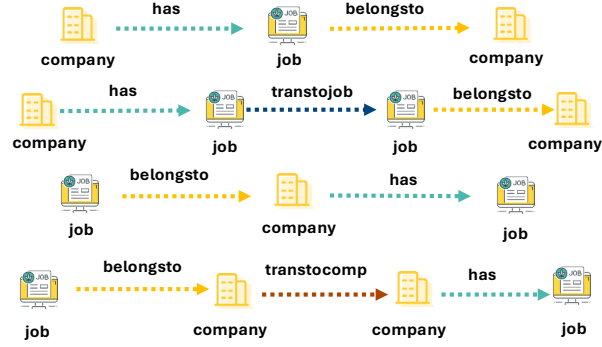


Fig. 2. The 4 meta-paths involved in career KG entity representation learning.

We also take the multi-head attention mechanism to make the learning of the updated feature $\mathbf{h}_i^{M'_p}$ focusing on various aspects as former works did[40][39]. Technically, we apply K independent self-attention mechanism for each node i and concatenate the resulted updated K features as the final updated feature for node i as shown in Equation 3.

$$\mathbf{h}_i^{M'_p} = \parallel_{k=1}^K \sigma \left(\sum_{j \in N_{M_p}(i)} \alpha_{ij}^{M_p^k} \mathbf{h}_j \mathbf{W}_{M_p}^{kT} \right) \in \mathbb{R}^{1 \times KF'}. \quad (3)$$

After going through the above MMP-GAT framework for all the meta-paths, we can now get multi-meta-paths semantic embeddings for both company and job nodes: $\mathbf{h}'_C = \{\mathbf{h}_C^{M'_p}\}_{p=1 \rightarrow N_{C_p}}$, $\mathbf{h}_C^{M'_p} \in \mathbb{R}^{C_N \times KF'}$ and $\mathbf{h}'_J = \{\mathbf{h}_J^{M'_p}\}_{p=1 \rightarrow N_{J_p}}$, $\mathbf{h}_J^{M'_p} \in \mathbb{R}^{J_N \times KF'}$ where C_N and J_N represents the number of company and job nodes, N_{C_p} and N_{J_p} are the number of meta-paths whose end nodes are both companies or both jobs respectively. Then we apply a multi-meta-paths semantic-level attention mechanism to combine each node's multi-semantic features represented by various meta-paths inspired by [42]. For each type of nodes set $V \in \{C, J\}$, we apply a shared linear projection layer with a mean pooling on all its nodes' meta-path-specific embeddings as shown in Equation 4 to get the importance scalar w_{M_p} for each meta-path.

$$w_{M_p} = \frac{1}{|V|} \sum_{i \in V} \mathbf{q}^T \tanh(\mathbf{W}_{\text{proj}} \mathbf{h}_i^{M'_p}). \quad (4)$$

Then we utilize a softmax layer for all w_{M_p} of meta-paths corresponding to current nodes set V to obtain the linear combination weights β_{M_p} for each meta-path as shown in Equation 5, where M_{V_p} is the number of meta-paths for current type nodes set V .

$$\beta_{M_p} = \frac{\exp(w_{M_p})}{\sum_{i=1}^{M_{V_p}} \exp(w_{M_i})}. \quad (5)$$

Finally, we get the multi-meta-paths semantic aware career KG entities' representations $\mathbf{H}_C$, $\mathbf{H}_J$ as a weighted combination of all meta-path-specific node embeddings as in Equation 6, where $\mathbf{H}_C \in \mathbb{R}^{C_N \times KF'}$ and $\mathbf{H}_J \in \mathbb{R}^{J_N \times KF'}$.

$$\mathbf{H}_C = \sum_{p=1}^{M_{C_p}} \beta_{M_p} \mathbf{h}_C^{M'_p}, \mathbf{H}_J = \sum_{p=1}^{M_{J_p}} \beta_{M_p} \mathbf{h}_J^{M'_p}. \quad (6)$$

We optimize the model using a multi-type link prediction objective. We choose link prediction graph task here because this aligns with our LLM's target: generating the most reasonable next career hop for each user can be regarded as predicting the most potential link between candidate entities. We pre-train MMP-GAT separately from LLM fine-tuning, as the two components target different modality interactions: learning structural patterns in the KG and understanding textual semantics.

4.3 Soft Prompt Construction

To effectively integrate a user's historical career information and leverage structural knowledge from the career KG, we construct soft prompts that fuse textual embeddings with graph-based node representations using a Bi-directional Cross-modality Interaction Adapter (BiCrossAda). Additionally, an entropy-based retrieval score guides the model to focus on the most relevant career entities while suppressing noise from less informative nodes. To realize this idea, we adopt a dual-modality encoding framework in which textual and graph information are encoded separately and later fused via BiCrossAda. For the textual modality, we use a pre-trained BERT model [7] to encode each historical (job, company) pair. The tokenized job titles and company names are fed separately into BERT, producing textual embeddings for historical job and company nodes: $\mathbf{T}_{\text{C}_{\text{his}}} \in \mathbb{R}^{n_{\text{his_c}} \times F_{\text{bert}}}$ and $\mathbf{T}_{\text{J}_{\text{his}}} \in \mathbb{R}^{n_{\text{his_j}} \times F_{\text{bert}}}$, where $n_{\text{his_c}}$ and $n_{\text{his_j}}$ denote the numbers of historical companies and jobs, respectively. $F_{\text{bert}} = 768$ is the fixed embedding dimension of the pre-trained BERT model. For the graph modality, we use the pre-trained MMP-GAT model to encode the retrieved sub-KG. Given the subgraph and its heterogeneous node features, MMP-GAT produces embeddings for all nodes, including the user's historical companies and jobs: $\mathbf{G}_{\text{C}_{\text{sg}}} \in \mathbb{R}^{n_{\text{sg_c}} \times KF'}$ and $\mathbf{G}_{\text{J}_{\text{sg}}} \in \mathbb{R}^{n_{\text{sg_j}} \times KF'}$, where $n_{\text{sg_c}}$ and $n_{\text{sg_j}}$ are the numbers of company and job nodes in the subgraph, and KF' denotes the output embedding dimension of MMP-GAT. We further introduce an entropy-oriented retrieval score that enables the model to automatically assign higher attention to informative nodes and lower attention to generic or biased ones. This mechanism encourages the retrieved subgraph to better reflect the user's historical career trajectory. Details of the retrieval score are provided in Section 4.3.1. The resulting attention weights are then applied to the corresponding node embeddings to obtain the final graph-modal representations. Both the textual and graph encoders are pre-trained and kept frozen during downstream training.

4.3.1 TF-IDF Inspired Entropy Based Retrieval Score.

Though we extract career sub-KG for each user as introduced in Section 4.1, we admit that this retrieval pipeline may involve nodes which are biased towards generic, uninformative career entities because the 2-hop graph searching tends to return high-degree nodes, for example, the well-known companies or popular jobs, which will reduce the sub graph's coverage for all the user-specific historical career information. Our goal is therefore to design a "retrieval score" that guides the model to assign higher attention to nodes most relevant to the user's historical career entities—the query nodes—and to ensure sufficient coverage of their associated information. We draw inspiration from the classical Term Frequency–Inverse Document Frequency (TF–IDF) measure [25], which quantifies the importance of a term to a document within a corpus. TF–IDF is computed as the product of the term frequency (TF) within the document and the inverse document frequency (IDF) across the corpus, as shown in Equation 7, where N is the total number of documents and $df(t)$ denotes the number of documents containing term t . A higher TF–IDF score indicates that term t is more informative for document d .

$$\begin{aligned} \text{TF-IDF}(t, d) &= \text{TF}(t, d) \text{IDF}(t), \\ \text{TF}(t, d) &= \frac{\text{num of } t \text{ in } d}{\text{num of all words in } d}, \\ \text{IDF}(t) &= \log\left(\frac{N}{df(t)}\right). \end{aligned} \tag{7}$$

In our setting, we aim to place higher attention on retrieved nodes that are more relevant to the user's historical job and company nodes. Therefore, it is natural to draw an analogy between retrieving relevant nodes given a user's historical career entities and retrieving relevant documents given query terms in the information-retrieval domain. Motivated by this analogy, we design a learnable retrieval score inspired by TF–IDF, as shown in Equation 7. We treat historical query nodes as query terms t and the candidate nodes in the subgraph as documents in a corpus. In TF–IDF, the term-frequency component $\text{TF}(t, d)$ measures how important a term t is to a document d . A higher $\text{TF}(t, d)$ indicates that term t is more important for document d . Analogously, we use the cosine similarity between the embeddings of query nodes and candidate nodes—derived from the pre-trained MMP-GAT—as the first component of our retrieval score. The second component, $\text{IDF}(t)$, measures how informative a term is across the entire corpus, and is inversely related to the frequency with which the term appears in documents. In the deep-retrieval setting,

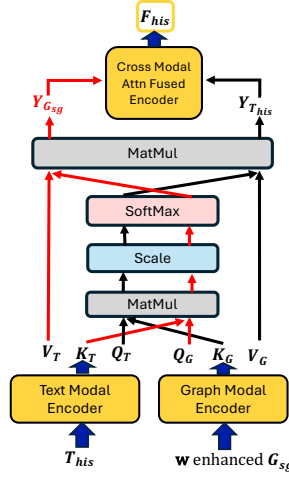


Fig. 3. Bi-directional Cross-modality Interaction Adapter

entropy can be used as a surrogate for IDF [22]. Specifically, for a historical query node $\mathbf{h} \in \mathbb{R}^{KF'}$ and a candidate node $\mathbf{n} \in \mathbb{R}^{KF'}$, we consider the multinomial distribution $\{\frac{\frac{1}{2}(1+\cos(\mathbf{h},\mathbf{n}))}{\sum_{n' \in G_{sg}} \frac{1}{2}(1+\cos(\mathbf{h},\mathbf{n}'))} : n \in G_{sg}\}$, where G_{sg} denotes the retrieved subgraph introduced in Section 4.1. This distribution has an entropy: $H(\mathbf{h})$. A large $H(\mathbf{h})$ corresponds to a query term with low IDF: the query node $\mathbf{h}$ exhibits no strong preference for any candidate node in G_{sg} . Consequently, the influence of $\mathbf{h}$ on its similarity scores with candidate nodes should be down-weighted, analogous to how TF-IDF down-weights terms with low IDF. We achieve this effect using the score defined in Equation 8, where $\bar{H}$ denotes the average entropy across all historical query nodes. Analogous to TF-IDF, a higher $s_{h,n}$ indicates that query node h is more relevant to candidate node n .

$$s_{h,n} = \frac{1}{2}(1 + \cos(\mathbf{h}, \mathbf{n}))\sigma(\bar{H} - H(\mathbf{h})). \quad (8)$$

We design the “retrieval score” to operate over the entire subgraph, with the goal of improving its coverage of user-specific historical career entities. By assigning a learnable weight to each node, the model can automatically learn to attend to nodes with different importance levels, rather than relying on a hard selection of a subset of nodes. We quantify how well the weighted subgraph covers a historical query node $\mathbf{h}$ using a soft maximization function, as defined in Equation 9. Here, $\mathbf{W}_{G_{sg}} \in \mathbb{R}^{(n_{sg,c} + n_{sg,j}) \times 1}$ denotes the learnable weight vector associated with all nodes in the current subgraph G_{sg} .

$$\mathbf{smax}(s_{h,n} : \text{for all } n \in G_{sg}) = \log\{\mathbf{W}_{G_{sg}}^T [\exp(s_{h,n})]_{n \in G_{sg}}\}. \quad (9)$$

Finally, the overall retrieval score is obtained by summing the coverage scores of all historical query nodes under the weighted subgraph, as shown in Equation 10.

$$\mathcal{L}_{RS} = \sum_{\mathbf{h} \in H_q} \mathbf{SMAX}(s_{h,n} : \text{for all } n \in G_{sg}). \quad (10)$$

4.3.2 Bi-directional Cross-modality Interaction Adapter: BiCrossAda.

Integrating LLMs with GNNs has become an active research direction for enhancing graph understanding and mitigating hallucinations in LLM outputs. Most existing approaches encode textual inputs and graph inputs separately, and then concatenate the two modalities as the final LLM prompt to explicitly incorporate structural information from the graph [13][29]. However, we argue that the simple concatenation operation ignores the complex and essential interactions between the two different modalities, which will prohibit LLM fully understand the structured knowledge provided by the retrieved subgraph given the corresponding text query. Motivated by this, we propose a bi-directional cross-modality interaction adapter, BiCrossAda, which constructs the final soft prompts for all historical career

entities by modeling the interactions between their textual and graph modalities. BiCrossAda is further equipped with a modality-attention-based fusion encoder, as illustrated in Figure 3.

We apply separate linear projection layers to the textual embeddings and the $\mathbf{W}_{G_{sg}}$ enhanced graph embeddings to get the query, key and value representations for the two modalities: $\mathbf{Q}_T, \mathbf{K}_T, \mathbf{V}_T \in \mathbb{R}^{n_{his} \times h_{cross}}$ and $\mathbf{Q}_G, \mathbf{K}_G, \mathbf{V}_G \in \mathbb{R}^{n_{sg} \times h_{cross}}$. Here, n_{his} denotes the number of a user's historical career entities (jobs or companies) and n_{sg} denotes the total number of nodes in the retrieved subgraph. h_{cross} is the dimensionality of the projected query, key and value embeddings for each modality. We then use scaled dot-product attention [39] to model bi-directional cross-modality interactions, where the query comes from one modality and the key-value pairs from the other. This produces updated graph embeddings $\mathbf{Y}_{G_{sg}} \in \mathbb{R}^{n_{sg} \times h_{cross}}$ and updated textual embeddings $\mathbf{Y}_{T_{his}} \in \mathbb{R}^{n_{his} \times h_{cross}}$, as shown in Equations 11 and 12.

$$\mathbf{Y}_{G_{sg}} = softmax(\frac{\mathbf{Q}_G \mathbf{K}_T^T}{\sqrt{h_{cross}}}) \mathbf{V}_T. \quad (11)$$

$$\mathbf{Y}_{T_{his}} = softmax(\frac{\mathbf{Q}_T \mathbf{K}_G^T}{\sqrt{h_{cross}}}) \mathbf{V}_G. \quad (12)$$

We then fuse each entity's textual and graph embeddings using a Cross-Modal Attention Fusion Encoder, as illustrated in Figure 3, through a weighted-sum operation, where modality-specific weights are learned through a self-attention mechanism by projecting each entity's textual and graph embeddings into modality-wise attention scores, $\mathbf{A}_T, \mathbf{A}_G \in \mathbb{R}^{n_{his} \times 1}$ as shown in Equation 13 to 15, where $\mathbf{W}_{attn} \in \mathbb{R}^{h_{cross} \times 1}$ is the shared learnable weight vector used in the projection layers.

$$\mathbf{Y}_{G_{sg}} \xrightarrow{\text{filter}} \mathbf{Y}_{G_{his}} \in \mathbb{R}^{n_{his} \times h_{cross}}, \quad (13)$$

$$\mathbf{A}_T, \mathbf{A}_G = split(softmax([\mathbf{Y}_{T_{his}} \mathbf{W}_{attn}] || [\mathbf{Y}_{G_{his}} \mathbf{W}_{attn}])), \quad (14)$$

$$\mathbf{E}_{fused} = \mathbf{A}_T \cdot \mathbf{Y}_{T_{his}} + \mathbf{A}_G \cdot \mathbf{Y}_{G_{his}}. \quad (15)$$

We apply the BiCrossAda module independently to the textual and graph embeddings of historical job and company entities, yielding the fused representations $\mathbf{E}_{C_{fused}} \in \mathbb{R}^{n_{his,c} \times h_{cross}}$ and $\mathbf{E}_{J_{fused}} \in \mathbb{R}^{n_{his,j} \times h_{cross}}$ for companies and jobs, respectively. Finally, we apply a shared linear projection layer to $\mathbf{E}_{C_{fused}}$ and $\mathbf{E}_{J_{fused}}$ to align them with the token embedding size of the backbone LLM. This produces the soft prompts for the user's historical jobs and companies: $\mathbf{P}_{soft_J} \in \mathbb{R}^{n_{his,j} \times F_{Llama}}$ and $\mathbf{P}_{soft_C} \in \mathbb{R}^{n_{his,c} \times F_{Llama}}$, where F_{Llama} denotes the fixed embedding dimension (4096) of the Llama-3.1-8B backbone.

4.3.3 KG Enhanced Prompts Construction.

We use Llama-3.1-8b [11] as our backbone LLM. We follow the official instruct prompt template¹ provided on the model's website. The Llama instruct prompt supports three roles: system, user, and assistant. The **system** role sets the context in which to interact with the LLM. In our setting, the system role specifies that the LLM acts as a career path planner assisting users in identifying potential job opportunities. The **user** role contains the inputs, instructions and queries provided to the model. We provide the user's historical career experience, educational background, and a set of candidate jobs as input to the **user** role, enabling the LLM to form a comprehensive view of the user's professional profile. The historical career experience consists of job title, company name, start and end dates, location, and job description for each career stage. The educational background includes the institution name, major, and start and end dates. The candidate jobs set contains 20 "(job title, company name)" pairs, including one ground-truth next career hop and 19 negative samples selected from the candidate pool that share the same company type as the ground-truth next company. The **assistant** role represents the response generated by the LLM based on the system and user prompts. We additionally introduce two special tokens into the tokenizer, $\langle |HisJobEmb| \rangle$ and $\langle |HisCompEmb| \rangle$, which serve as placeholders for the learned soft prompts of historical job and company entities, $\mathbf{P}_{soft_J}$ and $\mathbf{P}_{soft_C}$. An example of the final KG enhanced prompt are shown in Figure 4.

¹https://www.llama.com/docs/model-cards-and-prompt-formats/llama3_1/#model-card

<|begin_of_text|><|start_header_id|>system<|end_header_id|>

Cutting Knowledge Date: December 2023

Today Date: 26 Jul 2024

You are a career path planner tasked with helping an individual find new job opportunities.

<|eot_id|><|start_header_id|>user<|end_header_id|>

You have access to this person's historical career descriptions and education background. Based on this information, your goal is to identify the most suitable next job from a set of candidate positions. The candidate job set is presented as a list of tuples, where the first element in each tuple is the job title and the second element is the corresponding company name.

The person's historical careers description:

- Career 1: **job title**: Lead Network Controller<|HisJobEmb|>; **company name**: SITA<|HisCompEmb|>; **start time**: 1995/3; **end time**: 1998/4; **job address**: Greater Atlanta Area, United States; **job content**: Regional support to Centralized Help Desks, Troubleshooting network, systems, and protocol analysis
- Career 2: **job title**: Network Analyst Global Network Control Support<|HisJobEmb|>; **company name**: SITA/Equant (Joint Venture)<|HisCompEmb|>; **start time**: 1998/4; **end time**: 2001/1; **job address**: Greater Atlanta Area, United States; **job content**: Global Third Level support to Network Help Desks.
- Career 3: **job title**: System Administrator Messaging Network Support<|HisJobEmb|>; **company name**: SITA<|HisCompEmb|>; **start time**: 2001/1; **end time**: 2002/4; **job address**: Greater Atlanta Area, United States; **job content**: Second Level Support of Messaging Infrastructure.
- Career 4: **job title**: Customer Service Manager<|HisJobEmb|>; **company name**: orange-business-services<|HisCompEmb|>; **start time**: 2002/4; **end time**: 2018/11; **job address**: Atlanta Georgia; **job content**: Champion for the overall Customer Experience, Operations Excellence, Main point of Customer contact for all Customer operational issues.

The person's education background:

- Education 1: **school name**: Lamar County Comprehensive High School; **major**: None; **start time**: 1979; **end time**: 1982.
- Education 2: **school name**: Southern Crescent Technical College; **major**: Cisco CCNA/CCNP; **start time**: 2001 年; **end time**: 2005 年.

The candidate jobs set:

[('Customer Service Director', 'Orange Business Services'), ('Senior Visualization Engineer', 'TomTom'), ('Senior Manager - Operations', 'Verizon'), ('Service Desk Analyst', 'Unitas Global'), ('Licensed Massage Therapist', 'Self Employed'), ('Writer', 'Self-Employed'), ('Manager', 'Verizon'), ('Analista programador'), ('KEYLAND SISTEMAS DE GESTION'), ('Project Consultant', 'Procensus, SA'), ('IT Consulting Manager', 'CGI')].

<|eot_id|><|start_header_id|>assistant<|end_header_id|>

Fig. 4. The KG Enhanced Prompt Design.

4.4 LLM Parameter Efficient Fine-tuning

To efficiently adapt a general-purpose LLM to job-mobility prediction without incurring the prohibitive cost of full fine-tuning, we perform supervised instruction tuning and employ QLoRA to preserve performance while significantly reducing memory footprint. Specifically, we adopt a supervised instruction-tuning strategy that minimizes the autoregressive loss between model predictions and ground-truth outputs, thereby aligning the LLM with the job-mobility generation task. We then employ QLoRA [6] to perform parameter-efficient fine-tuning of the backbone LLM, achieving lower memory consumption than LoRA [26] while incurring almost no performance degradation. QLoRA first quantizes the model to 4-bits and then applies LoRA-based training, which freezes the pre-trained model parameters and introduces trainable low-rank decomposition matrices into each Transformer layer to enable light-weight fine-tuning [15]. Accordingly, the autoregressive loss is computed as follows:

$$\mathcal{L}_{\mathcal{AR}} = \max_{\Theta_{qL}} \sum_{(e_x, y) \in \mathcal{T}} \sum_{t=1}^{|y|} \log(P_{\Theta + \Theta_{qL}}(y_t | e_x, y_{<t})). \quad (16)$$

Here, Θ_{qL} denotes the QLoRA parameters, while Θ represents the frozen parameters of the pre-trained LLM; e_x and y correspond to the KG-enhanced prompt and its ground-truth response, respectively. $\mathcal{T}$ denotes the training set, and y_t is the t -th token in the answer. Our final objective combines the autoregressive loss in Equation 16 with the retrieval score in Equation 10, as shown in Equation 17, where λ is a balancing hyperparameter.

$$\mathcal{L} = \mathcal{L}_{\mathcal{AR}} + \lambda \mathcal{L}_{\mathcal{RS}}. \quad (17)$$

5 Experiments

In this section, we present experimental details based on two real-world datasets to validate the proposed framework. The experiments include comparative studies, evaluations of SCP's generalizability across different backbone LLMs, ablation studies, analyses of data efficiency and robustness, parameter sensitivity analysis, and an illustrative case study demonstrating the effectiveness of our framework.

5.1 Experimental Settings

5.1.1 Datasets.

We conduct experiments on two real-world datasets, FIN and IT, which are collected from LinkedIn, one of the world's largest professional recruitment networking platform. The FIN and IT datasets include LinkedIn users from the finance and information technology industries, respectively, whose career histories and educational backgrounds are complete and contain between two and ten career records. The cut-off date for data collection is December 1, 2019. For each user, we treat the first $n - 1$ recorded careers as historical input and aim to predict the n -th career in the form of a $(job, company)$ pair. In addition, we extract textual information from each user's educational background to obtain a more comprehensive representation of the individual. The candidate jobs set consists of one ground-truth pair (i.e., the user's n -th career) and 19 randomly selected negative pairs drawn from the same career domain. Dataset statistics are summarized in Table 2.

Table 2. Statistics of Datasets

Attributes	FIN	IT
# Train Users	14,206	18,409
# Val Users	1,785	2,280
# Test Users	1,830	2,242
# Average Length of Career Path	2.78	2.77
# Average Number of Records of Educations	1.72	1.70

5.1.2 Compared Baseline Models.

We compare four categories of baseline models: traditional sequential models, GNN-based models, enhanced small-scale LLMs, and large-scale foundation LLMs. The overall comparison results are reported in Table 3.

Traditional Sequential Models.

- **LSTM**: uses a standard LSTM to model the dynamics of an individual's career path without incorporating profile-level contextual information [14].
- **NEMO**: a contextual LSTM-based model that jointly integrates profile information and career path dynamics to predict the next company or position of a user [24].
- **HCPNN**: a hierarchical career-path-aware neural network that models individual job mobility by predicting both the next employer and the corresponding job duration [27].

GNN-Based Models.

- **Metapath2Vec**: embeds nodes in heterogeneous graphs by combining meta-path-guided random walks with a heterogeneous Skip-Gram model [8].
- **HAN**: applies hierarchical attention mechanisms (node-level and semantic-level) over meta-path-based heterogeneous graphs to learn semantically and structurally aware node representations [42].
- **HGT**: a heterogeneous graph neural architecture that employs node- and edge-type-dependent attention to produce type-specific embeddings, enabling effective modeling of large-scale heterogeneous graphs [16].

Enhanced Small-Scale LLMs.

- **Fine-Tuned LLaMA via QLoRA**: fine-tunes the backbone LLM (Llama-3.1-8B) using QLoRA with carefully designed job-mobility prediction prompts.
- **G-Retriever**: a retrieval-augmented generation framework for textual graphs, which enhances graph understanding through soft prompting and can be fine-tuned for GraphQA tasks [13].
- **GraphToken**: a parameter-efficient encoding method that incorporates structured data into LLM prompts via learned graph representations [29].

Large-Scale Foundation LLMs.

- **DeepSeek-V3.2-Exp**: a 685-billion-parameter LLM that introduces a novel sparse attention mechanism to significantly improve long-context training and inference efficiency while maintaining the output quality of its predecessor. [5]

5.1.3 Evaluation Metrics.

To evaluate job mobility prediction accuracy, we adopt **HitRatio@1**, **HitRatio@3** and **HitRatio@5**, following common practice in prior job mobility prediction studies. **HitRatio@K** measures whether the ground-truth company/job appears in the Top-K predicted results, as defined in Equation 18. In addition, since our prediction framework follows a generative paradigm that may produce invalid outputs (e.g., nonsensical tokens or items outside the candidate set), we introduce an additional metric, **ValidRatio**. ValidRatio quantifies the proportion of valid responses (i.e., items drawn from the candidate set) across all predictions, serving as an indicator of the model’s instruction-following capability, as defined in Equation 19.

$$\text{HitRatio@K} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y_i \in \text{Top-K}(y_i^{\text{pred}})]. \quad (18)$$

$$\text{ValidRatio} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{Top-1}(y_i^{\text{pred}}) \in \text{CandidateSet}]. \quad (19)$$

5.2 Implementation Details

All experiments were conducted on two NVIDIA RTX A6000 GPUs. We use Llama-3.1-8b [11] as the backbone LLM. AdamW is adopted as the optimizer with a learning rate of 5×10^{-4} for both datasets. For fairness, all experiments involving LLM fine-tuning were trained for three epochs with an early stopping strategy. Details of model parameters on both datasets can be found in Table 4. The model parameters mainly fall into three components of the overall pipeline: (1) GNN-related, including the GNN hidden dimension, number of attention heads, and dropout rate; (2) BiCrossAda-related, including the cross-modal embedding dimension and the hidden dimension of the fusion encoder; (3) Retrieval-score-related, including the loss balancing hyperparameter.

5.3 Results

5.3.1 Performance Comparison with Baseline Models.

The results of the comparative experiments with all baseline models are summarized in Table 3. For each dataset, the best results are highlighted in **bold**, and the second-best results are underline. Our model achieves the best performance on both datasets, each containing fewer than 20,000 career trajectories. In contrast, the inferior performance of traditional job mobility prediction baselines can largely be attributed to insufficient training data. As reported in the original papers, the dataset sizes used to train NEMO [24] and HCPNN [27] exceed 1M and 414,266 samples, respectively. We further compare our method with representative GNN-based baselines, including Metapath2Vec [8], HAN [42], and HGT [16]. These models consistently outperform traditional sequential approaches by leveraging graph structural information, with HGT achieving the strongest performance among them due to its heterogeneous attention mechanism. However, all GNN-based methods remain inferior to our approach, indicating that graph structure alone is insufficient without joint semantic reasoning enabled by LLMs. Furthermore, compared with fine-tuning LLaMA-3.1-8B using QLoRA with carefully designed prompts—a strong baseline for aligning pretrained LLMs with domain-specific downstream tasks [2]—our model achieves improvements of **11.9%** and **8.6%** in Hit@1, 6.06% and 0.6% in Hit@3, and 4.17% and 0.6% in Hit@5 on the IT and FIN datasets, respectively. These consistent performance gains further validate the effectiveness of incorporating knowledge retrieved from the career knowledge graph. Moreover, we compare our model with two representative state-of-the-art approaches for LLM-enhanced graph understanding via soft prompting: G-Retriever[13] and GraphToken[29]. G-retriever formulates graph retrieval as a PCST optimization problem with a hard sub-graph selection process. In contrast to G-Retriever, we introduce a learnable retrieval score that allows the model to adaptively allocate attention to different entities within each retrieved sub-graph. GraphToken employs a graph encoder to transform the input graph into a fixed number of

Table 3. Performance comparison across models and datasets on company and job prediction tasks.

Dataset	Model	Hit@1	Hit@3	Hit@5	Valid%
IT	LSTM	0.0477 ± 0.0061	0.1734 ± 0.0098	0.3008 ± 0.0138	1.0000
	NEMO	0.0434 ± 0.0029	0.1553 ± 0.0100	0.2619 ± 0.0166	1.0000
	HCPNN	0.0329 ± 0.0035	0.1629 ± 0.0111	0.2967 ± 0.0162	1.0000
	Metapath2Vec	0.3358 ± 0.0037	0.4575 ± 0.0040	0.5468 ± 0.0051	1.0000
	HAN	0.0704 ± 0.0031	0.1933 ± 0.0096	0.3073 ± 0.0135	1.0000
	HGT	0.3990 ± 0.0591	0.5589 ± 0.0631	0.6527 ± 0.0667	1.0000
	LLaMA-3.1-8b + qLoRA	0.5357 ± 0.0115	0.7603 ± 0.0030	<u>0.8479 ± 0.0037</u>	0.9243 ± 0.0059
	G-Retriever	<u>0.6012 ± 0.0113</u>	<u>0.7885 ± 0.0045</u>	0.8236 ± 0.0067	<u>0.9979 ± 0.0007</u>
	GraphToken	0.4784 ± 0.0068	0.6786 ± 0.0095	0.7376 ± 0.0074	0.9851 ± 0.0023
	DeepSeek-V3.2-Exp (685b)	0.4505	0.5071	0.6677	0.9153
	SCP (Ours)	0.6115 ± 0.0071	0.8082 ± 0.0059	0.8850 ± 0.0048	0.9986 ± 0.0006
FIN	LSTM	0.0501 ± 0.0042	0.1677 ± 0.0034	0.2745 ± 0.0046	1.0000
	NEMO	0.0457 ± 0.0008	0.1593 ± 0.0064	0.2725 ± 0.0122	1.0000
	HCPNN	0.0307 ± 0.0026	0.1421 ± 0.0062	0.2658 ± 0.0049	1.0000
	Metapath2Vec	0.4393 ± 0.0059	0.5502 ± 0.0046	0.6240 ± 0.0082	1.0000
	HAN	0.1198 ± 0.0052	0.2846 ± 0.0140	0.4148 ± 0.0027	1.0000
	HGT	0.4825 ± 0.0782	0.6165 ± 0.0872	0.6873 ± 0.0877	1.0000
	LLaMA-3.1-8b + qLoRA	0.6696 ± 0.0063	<u>0.8579 ± 0.0045</u>	<u>0.9083 ± 0.0017</u>	0.9472 ± 0.0031
	G-Retriever	<u>0.7050 ± 0.0140</u>	0.8327 ± 0.0109	0.8598 ± 0.0047	<u>0.9956 ± 0.0006</u>
	GraphToken	0.6222 ± 0.0163	0.7681 ± 0.0048	0.7998 ± 0.0055	0.9902 ± 0.0020
	DeepSeek-V3.2-Exp (685b)	0.5585	0.6033	0.7202	0.8929
	SCP (Ours)	0.7330 ± 0.0040	0.8777 ± 0.0023	0.9220 ± 0.0027	0.9986 ± 0.0007

token embeddings, which are prepended to the prompt tokens of a frozen pre-trained LLM. In contrast, our approach fine-tunes the backbone LLM using QLoRA. To ensure a fair comparison, we keep all GNN-related parameters consistent across the comparison experiments. Our model consistently outperforms both G-retriever and GraphToken, achieving improvements of **2.2%** and **12.8%** on the FIN dataset, and **1.8%** and **19.2%** on the IT dataset, respectively. We also include a large-scale foundation LLM, DeepSeek-V3.2-Exp (685B parameters), as an additional baseline. Despite its substantially larger model size, it underperforms SCP on both datasets, suggesting that scaling model capacity alone is insufficient for job mobility prediction. In contrast, SCP benefits from explicit integration of career knowledge graph structure, demonstrating the importance of domain-aware graph-LLM alignment over pure model scaling. These results further demonstrate the effectiveness and robustness of the proposed pipeline.

5.3.2 SCP with Different Backbone LLMs.

Evaluating the performance of the proposed SCP framework across different backbone LLMs is crucial for assessing its generalizability and adaptability. In this set of experiments, we select three widely used LLMs pre-trained by leading AI companies, which have demonstrated strong performance across a variety of tasks: **Mistral_7B_Instruct** [19], **Qwen2-7B-Instruct** [1] and **Llama-3.1-8B** [11]. By evaluating the framework with different backbone LLMs,

Table 4. Model Parameters

Attributes	FIN	IT
batch size	64	64
GNN hidden dim	512	256
GNN num heads	8	4
GNN dropout	0.2	0.4
Cross-Modal embedding dim	1024	512
fusion encoder hidden dim	2048	1024
loss balance parameter	1	1.4

we assess how effectively it integrates external knowledge from the career knowledge graph and adapts to the characteristics of each LLM for the job mobility prediction task. As shown in Table 5, Llama-3.1-8B achieves the best alignment with the proposed pipeline, obtaining the highest Hit@1, Hit@3, and Hit@5 scores across both datasets. Mistral-7B-Instruct and Qwen2-7B-Instruct also perform competitively, with Qwen2-7B-Instruct achieving the same Hit@3 score as Llama-3.1-8B on FIN dataset. These comparison results highlight the robustness of SCP in leveraging diverse LLM architectures and demonstrate its ability to consistently improve job mobility prediction accuracy. Furthermore, these evaluations provide insights into the framework’s compatibility with different LLMs, helping identify suitable backbone models for job mobility prediction.

Table 5. Results Table for SCP with Different LLMs

SCP_w_diff_LLM	IT				FIN			
	Hit@1	Hit@3	Hit@5	Valid%	Hit@1	Hit@3	Hit@5	Valid%
SCP_w_Mistral_7B_Instruct	0.5785	0.7935	0.8787	0.9969	0.7011	0.8448	0.9055	0.9973
SCP_w_Qwen2_7b_instruct	0.5825	0.7939	0.8524	0.9929	0.7000	0.8689	0.9016	0.9956
SCP_w_Meta_Llama_3.1_8B_Instruct	0.5914	0.8122	0.8827	0.9973	0.7169	0.8689	0.9115	0.9956

5.3.3 Ablation Study.

In the ablation study, we evaluate the individual contributions of three key components in our pipeline by removing each component in turn on both datasets, resulting in three variants: ours_w/o_MMP-GAT, ours_w/o_Retrieval Score and ours_w/o_BiCrossAda. The results are shown in Figure 5. We observe consistent performance drops when any of these components is removed, indicating that each module plays a critical role in the overall framework.

5.3.4 Data Efficiency and Robustness Analysis.

To evaluate the robustness and data efficiency of the proposed SCP framework, we train the model using different proportions of the training data—20%, 40%, 60%, and 80% of the full dataset—while keeping all other hyperparameters fixed (with the random seed set to 1 for all experiments). The performance trends on the FIN and IT datasets are shown in Figure 6.

As illustrated in Figure 6, SCP maintains stable performance across different training data proportions on both datasets. Across all evaluation metrics (Hit@1, Hit@3, Hit@5, and Valid%), the model exhibits only minor performance fluctuations, demonstrating strong robustness to variations in training data volume. Even when trained with only 20% of the data, SCP achieves competitive performance (e.g., Hit@1 above 0.57 on IT and 0.69 on FIN), indicating its effectiveness in low-resource settings. The Valid% metric consistently remains above 0.99 across all settings, confirming that the model preserves reliable instruction following and output validity regardless of data scale. Overall, these results highlight the framework’s strong data efficiency and robust performance across domains.

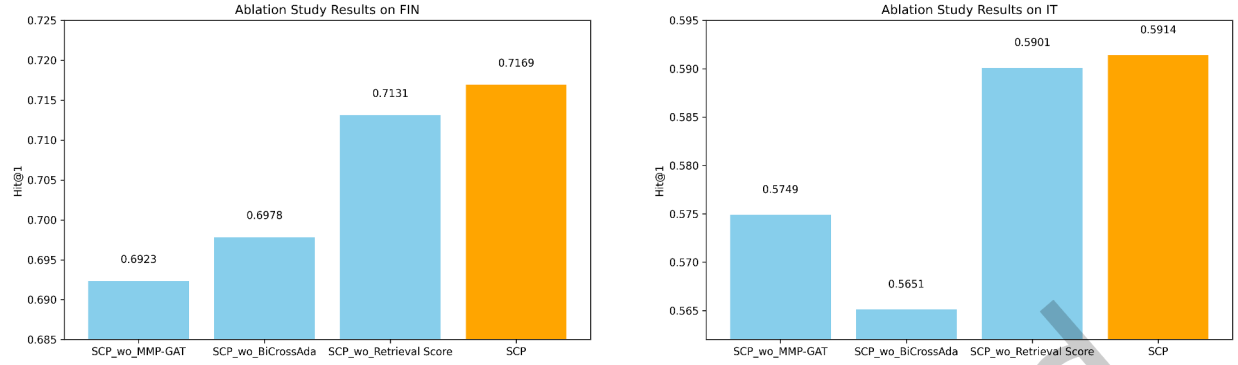


Fig. 5. Ablation study on FIN and IT datasets.

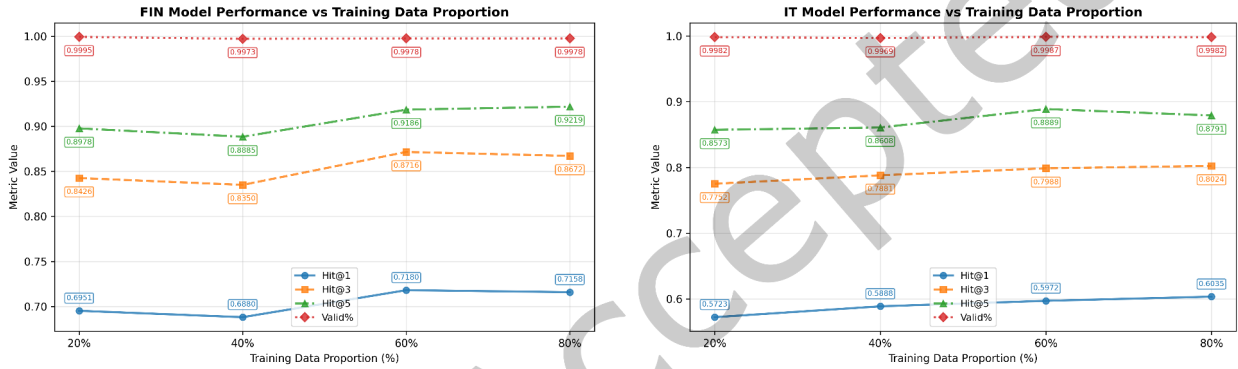


Fig. 6. Model performance on FIN and IT dataset with different training data proportions.

5.3.5 Parameter Experiments.

Here, we conduct parameter experiments on two key parameters in our pipeline: a critical hyperparameter, the **loss balance parameter** λ , and a model parameter, the **cross-modality embedding dimension** h_{cross} . The experimental results are reported in Table 6 and 7, where λ ranges from 0.4 to 1.0 and h_{cross} varies from 128 to 1024. For the **loss balance parameter** λ , larger values correspond to a greater influence of the retrieval score on next career hop prediction. On the FIN dataset, the optimal value of λ within the examined range is 1.0, whereas the optimal λ for the IT dataset is 0.8. Overall, increasing the influence of the retrieval score benefits job mobility prediction by encouraging the model to attend more to user-specific historical career entities within the retrieved career sub-KG. The **cross-modality embedding dimension** h_{cross} denotes the dimensionality of the query, key, and value vectors in the BiCrossAda module for each modality. The optimal h_{cross} is 256 for the FIN dataset and 128 for the IT dataset. The results indicate that increasing h_{cross} does not necessarily lead to better performance, potentially due to overfitting. A well-chosen h_{cross} helps the model more effectively capture cross-modality interactions between text and graph representations, thereby generating more informative soft prompts for the LLM and reducing hallucinations.

5.3.6 Case Study and Visualization.

To provide a more intuitive understanding of the effect of the retrieval score, we select one representative sample from each of the FIN and IT test datasets for which the next career pair is correctly generated, as shown in Figure 7. In each figure, the left panel presents the current sample's historical jobs and companies, along with the ground-truth next career pair that the LLM is expected to generate. The right panel visualizes the retrieved user-specific 2-hop sub-KG, constructed by selecting the top 20 most correlated career entities learned by the model, with only connected nodes and their corresponding edges retained. Green blocks denote job entities, while yellow blocks denote company entities. In the FIN case study, the retrieved sub-KG reveals a clear transition pattern from lower-level managerial

Table 6. Effect of different loss balance parameters

Loss Balance Parameters	FIN		IT	
	Hit@1	Valid%	Hit@1	Valid%
0.4	0.7038	0.9989	0.5629	0.9982
0.6	0.6989	0.9973	0.5839	0.9964
0.8	0.6956	0.9973	0.5879	0.9978
1.0	0.7169	0.9956	0.5745	0.9964

Table 7. Effect of different cross embed dimensions

Cross Embed Dimension	FIN		IT	
	Hit@1	Valid%	Hit@1	Valid%
128	0.7060	0.9956	0.5972	0.9973
256	0.7219	0.9962	0.5870	0.9955
512	0.7071	0.9989	0.5914	0.9973
1024	0.7169	0.9956	0.5897	0.9955

roles to more advanced managerial positions within small financial companies (e.g., a transition from branch manager to project manager). This pattern aligns well with the sample's observed career trajectory, which progresses from a general managerial role to a branch manager position. Similarly, in the IT case study, the retrieved sub-KG captures a job-hopping pattern from technical roles (e.g., software engineer) to consulting positions in professional service companies (e.g., LAN expert), which also closely matches the sample's career development path. These case studies demonstrate that the proposed entropy-based retrieval score effectively retrieves structured knowledge aligned with users' historical career transition patterns, thereby improving the robustness and accuracy of next career hop prediction.

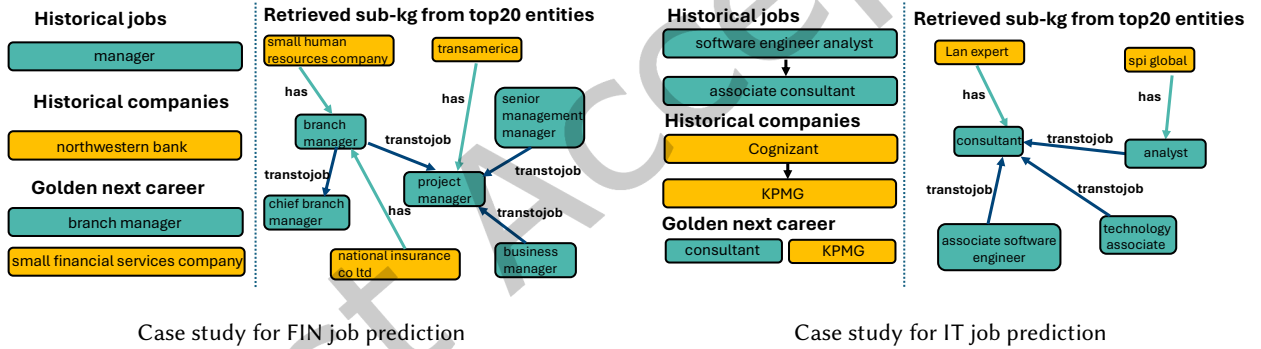


Fig. 7. Case Study

6 Conclusion

In this paper, we investigate the job mobility prediction problem by integrating large language models (LLMs) with a career knowledge graph (KG) through soft prompting. We introduce MMP-GAT, a heterogeneous GNN pre-trained on sub-KGs via a self-supervised link prediction task, to generate graph-modal representations of career entities. To better exploit the retrieved sub-KGs, we propose a retrieval score that guides the model's attention across different entities. In addition, we employ a frozen BERT encoder to construct textual representations of career entities. Furthermore, we introduce BiCrossAda, a bi-directional cross-modality interaction adapter that enhances soft prompts by explicitly modeling interactions between textual and graph modalities. Extensive experiments on two real-world datasets demonstrate the effectiveness of the proposed framework.

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